Presentation Outline

Slide-by-slide outline and speaker notes for the final presentation

Project title	IntelliMetrics AI: A Governed Semantic Layer with TARA, an AI Assistant for Trusted Business Metrics
Document	Presentation Outline
Team members	Devi Priya, Priyanka, Akshay and Bhaskar
Guide	Dr. Asesh Kumar Tripathy
Institution	KL University
Academic year	2026-27
Date	24-Oct-2026 (IST)
Status	University project: academic capstone with synthetic demonstration data

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1 Purpose and timing

The final presentation tells the story in this order: the problem, the idea, how it works, proof that it works, and an honest account of what is missing. It is designed for **about 15 minutes of talking plus an 8-minute live demonstration** and questions. A PowerPoint file with the same slides and speaker notes is provided as `exports/08_Presentation.pptx`; the visuals are the diagrams, charts and screenshots generated for the reports.

Part	Slides	Suggested time
Problem and objectives	1–3	2 min
Solution and architecture	4–5	2 min
Core ideas: governance, safe formulas, conflicts, TARA	6–9	5 min
Indian context	10	1 min
Live demonstration	11	8 min
Evidence: testing and performance	12–13	2 min
Limitations, future scope, close	14–16	2 min

2 Tips for the speaker

1. Open with the GST example; it makes the problem concrete in two sentences.
2. Say “university project, synthetic data” early, and “no live AI key was used, no Docker run” before anyone asks.
3. Show, then explain: the live demo should come before the test statistics.
4. If the live demo fails, use the backup screenshots in the slide and the walkthrough table in the User Guide.
5. Keep one sentence ready for each number: *what was measured, on what, and what it does not show*.

3 Slide-by-slide outline

The deck `exports/08_Presentation.pptx` contains these slides, with the speaker notes below in the PowerPoint notes pane.

3.1 Slide 1: IntelliMetrics AI

- A governed semantic layer with TARA, an AI assistant for trusted business metrics
- Devi Priya, Priyanka, Akshay and Bhaskar
- Guide: Dr. Asesh Kumar Tripathy | KL University | AY 2026-27
- University project (academic capstone). Demonstration data is synthetic.

Speaker notes: Introduce yourself, the guide and the project name. Say at once that it is a university project with synthetic data.

3.2 Slide 2: The problem: one word, many numbers

- Sales: Revenue = taxable value of goods sold (excluding GST)
- Finance: Revenue = sales less refunds
- Marketing: Revenue = invoice value including GST
- Nobody notices until two reports disagree; changes are not recorded
- Generic AI chatbots answer confidently even when wrong

Speaker notes: Use the Indian GST example because the audience recognises it. Stress that nobody is lying; the same word means different calculations.

3.3 Slide 3: Objectives

- One governed live definition per business metric
- Detect conflicting definitions automatically and measure the detector
- Immutable version history with safe rollback
- Let non-technical users ask questions without risking wrong or unsafe queries
- Localise to India: INR (lakh/crore), April-March FY, GST, festivals, IST

Speaker notes: Map each objective to a result later in the talk.

3.4 Slide 4: The solution at a glance

- Datasets in, governed metrics and dashboards out
- Workflow: draft, review, approve, publish (exactly one live per metric)
- Conflict centre: assign, review, approve, resolve
- Formula Lab: safe, parameterised execution
- Ask TARA: plain-English questions answered from approved metrics
- 96 API endpoints | 25 pages | 4 roles | 2 isolated workspaces

Visual: `diagrams/14_use_case_roles.png`

Speaker notes: Give the one-minute product tour using the use-case diagram: four roles, each with a distinct job.

3.5 Slide 5: Architecture

- React + TypeScript single-page app
- FastAPI back end: RBAC on every route, workspace filter on every query
- PostgreSQL / SQLite for metadata, DuckDB + Parquet for analytics
- Celery + Redis for background ingestion (thread fallback)

Visual: `diagrams/01_high_level_architecture.png`

Speaker notes: Explain that analytics never touch the transactional database; DuckDB reads Parquet files produced by ingestion.

3.6 Slide 6: Governance: versions that cannot be changed

- Partial unique index: one live definition per metric key
- Version rows immutable: ORM guard + database trigger
- Rollback creates a NEW version; history is never rewritten
- Dashboards follow latest published or pinned versions

Visual: `diagrams/08_metric_state_machine.png`

Speaker notes: Say that the guarantee is in the database, so a bug in the application cannot break it. Verified on real PostgreSQL 18.

3.7 Slide 7: A formula engine that cannot be injected

- Lexer, recursive-descent parser, typed syntax tree
- Whitelisted functions; columns checked against the schema
- Every value is a bound parameter; division is zero-safe
- Limits: 2,000 characters, depth 40, 10,000 rows, 20 s timeout
- 25 hostile inputs tested: all rejected before execution

Visual: `diagrams/06_formula_compile_pipeline.png`

Speaker notes: No user text ever becomes SQL text. That is why injection is structurally impossible rather than filtered.

3.8 Slide 8: Automatic conflict detection

- Compares formula, aggregation, filters, unit, time window, dataset
- Severity score, side-by-side diff, recommended resolution
- 27 labelled cases: precision 1.00, recall 1.00
- Plain text comparison baseline: precision 0.55, recall 0.81

Visual: `diagrams/07_conflict_detection_flow.png`

Speaker notes: Be candid: the cases were written by me, so this shows the rules behave as designed and is not an independent benchmark.

3.9 Slide 9: Ask TARA: the AI never writes numbers

- Question becomes a plan; the plan is validated against approved metrics
- Same safe compiler runs it; the answer text is composed from rows
- Compare ranges, second dimension, share of total, default fallbacks
- Refuses when the data cannot answer

Visual: `diagrams/04_ask_tara_request_flow.png`

Speaker notes: Walk through the sequence diagram. Mention the offline deterministic mode and that no live AI key was used.

3.10 Slide 10: Built for India

- INR with lakh/crore grouping; compact forms ₹45 L, ₹1.2 Cr
- Financial year 1 April to 31 March; fiscal quarters
- GST: taxable value vs invoice value; slabs 0/5/12/18/28 %
- Festival windows: Diwali, Dussehra, Pongal, Eid, Holi, End-of-Season sale
- DD-MMM-YYYY dates in IST

Visual: `diagrams/12_fiscal_calendar_and_planner.png`

Speaker notes: Explain that 'last quarter' resolves to the last completed fiscal quarter, and 'this year' to the financial year to date.

3.11 Slide 11: Demonstration

- Reviewer: detect and resolve the Revenue conflict
- Analyst: Formula Lab, including a rejected injection
- Business user: Diwali comparison and UPI vs COD split
- Admin: audit log; other workspace: no cross-company data

Visual: `screenshots/light/41_ask_tara_empty.png`

Speaker notes: Run the live demo here (8 minutes). The screenshot is a backup if the demo fails.

3.12 Slide 12: Testing

- 325 automated tests, 0 failed
- pytest 136 (+ 136 on PostgreSQL 18), Vitest 40, Playwright 13
- Back-end statement coverage 82.8 %
- Security tests: injection, token forgery, CSRF, isolation, escalation
- Every page screenshotted in light and dark mode

Visual: `charts/tests_by_suite.png`

Speaker notes: Each row of the test-case table in the report is linked to a measured result, so the table cannot claim a pass that did not happen.

3.13 Slide 13: Measured performance

- Query engine median 6.37 ms, p95 12.77 ms on 62,000 rows
- Ingestion about 86,494 rows per second
- Dashboard load 167.2 ms on average
- Time-saving estimates rely on stated assumptions, not measurements

Visual: `charts/query_latency.png`

Speaker notes: Single-user numbers from one Windows machine. Say clearly that there was no load test.

3.14 Slide 14: Honest limitations

- Docker Compose written but never run (Docker not installed)
- No live AI provider tested; offline planner only
- Synthetic data; no user study; no external security audit
- Chromium only; no load test; per-process rate limiter
- Verified instead: PostgreSQL 18.2, Redis + Celery, MySQL 8.0

Speaker notes: Say these before the examiner asks. It builds trust.

3.15 Slide 15: Future scope and conclusion

- Evaluate with a real LLM on a held-out question set
- Redis-backed rate limiting; e-mail delivery; single sign-on
- Column-level security; connectors to real BI sources; user study
- One definition, one history, answers you can trace

Speaker notes: Close with the contribution: an integrated, tested, honest implementation.

3.16 Slide 16: Thank you. Questions?

- Devi Priya, Priyanka, Akshay and Bhaskar
- Guide: Dr. Asesh Kumar Tripathy
- Documents: Project Report, Concept Note, Architecture, Testing Report, User Guide, Deployment Guide, Viva Preparation

Speaker notes: Invite questions. The Viva Preparation document lists forty likely ones.